Strategies for preparing successful grant proposals

NYIT Faculty Development Day

John Zweck

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https://drjohnzweck.github.io

Workshop Panelists

- Joanne Donoghue (Long Island)
  - Director of Clinical Research & Professor
  - Department of Osteopathic Manipulative Medicine
  - jdonoghu@nyit.edu
  - https://site.nyit.edu/bio/jdonoghu
- Amanda Golden (Long Island)
  - Associate Professor of English
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  - https://site.nyit.edu/bio/agolde01
- Yanhua (Jennifer) Xie (NYITCOM)
  - Associate Professor of Biomedical and Anatomical Sciences
  - Jennifer.Xie@nyit.edu
  - https://site.nyit.edu/bio/jennifer.xie

Why Write Grant Proposals?

- Support NYIT's institutional push to R2 status
- To realize the research goals you are passionate about
- Facilitate research collaborations
- Raise your research profile
- Provide funding for
  - Graduate and Undergraduate Research Assistantships
  - Research Lab
  - Faculty Summer Salary
  - Travel to Conferences
  - Teaching relief

Forces you to develop a 3-5 year research program

Forces you to produce and deliver

What is your compelling big idea?

Your proposal should attempt to change how we think about a topic and excite the reader.

Answer Three Key Questions

1. What is the problem?
   - Strong research question
   - Between something "sure" and something truly innovative
2. Why is it important?
   - Who cares?
   - So what? What happens if you do this?
3. What is your key idea?
   - How is your approach innovative?
   - How are you going to do it?
   - Why are you the best person for the project?

Program managers & 1-page white paper

- Write a 1-page white paper on the 3 Key Questions
- Reach out to program manager
  - Email or set up phone call
  - Attached your white paper and a CV
  - Come with specific questions they can answer
  - Volunteer to be a reviewer

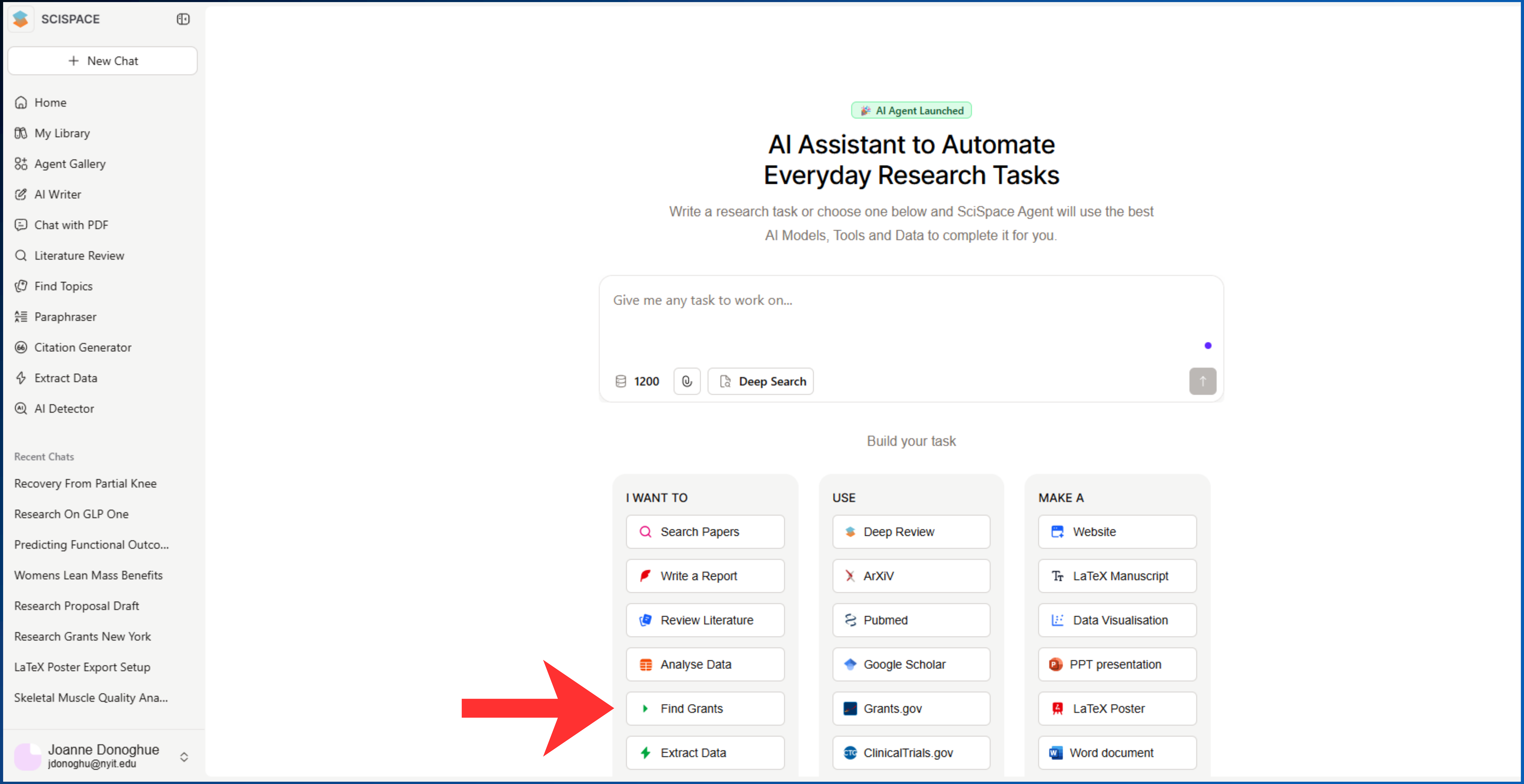
Artificial Intelligence

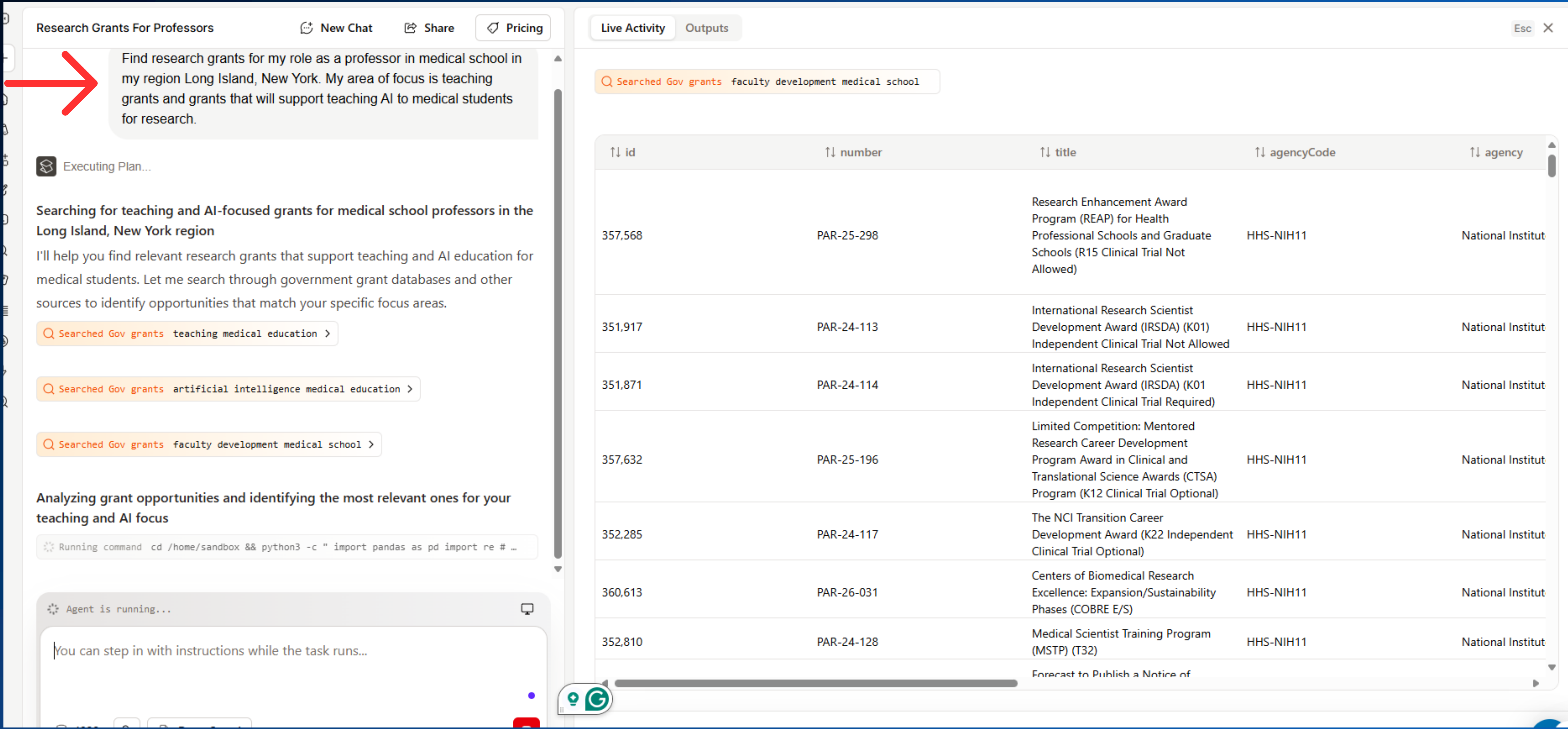
To find your grant

SCISPACE

ChatGPT

Joanne Donoghue PhD
Director of Clinical Research,
College of Osteopathic Medicine


SCISPACE
New Chat
Home
My Library
Agent Gallery
AI Writer
Chat with PDF
Literature Review
Find Topics
Paraphraser
Citation Generator
Extract Data
AI Detector
Recent Chats
Recovery From Partial Knee
Research On GLP One
Predicting Functional Outco...
Womens Lean Mass Benefits
Research Proposal Draft
Research Grants New York
LaTeX Poster Export Setup
Skeletal Muscle Quality Ana...
Joanne Donoghue
jdonoghu@nyit.edu
AI Agent Launched
AI Assistant to Automate
Everyday Research Tasks
Write a research task or choose one below and SciSpace Agent will use the best
AI Models, Tools and Data to complete it for you.
Give me any task to work on...
1200
Deep Search
Build your task
I WANT TO
Search Papers
Write a Report
Review Literature
Analyse Data
Find Grants
Extract Data
USE
Deep Review
ArXiV
Pubmed
Google Scholar
Grants.gov
ClinicalTrials.gov
MAKE A
Website
LaTeX Manuscript
Data Visualisation
PPT presentation
LaTeX Poster
Word document


Research Grants For Professors
New Chat
Share
Pricing
Find research grants for my role as a professor in medical school in my region Long Island, New York. My area of focus is teaching grants and grants that will support teaching AI to medical students for research.
Executing Plan...
Searching for teaching and AI-focused grants for medical school professors in the Long Island, New York region
I'll help you find relevant research grants that support teaching and AI education for medical students. Let me search through government grant databases and other sources to identify opportunities that match your specific focus areas.
Searched Gov grants teaching medical education
Searched Gov grants artificial intelligence medical education
Searched Gov grants faculty development medical school
Analyzing grant opportunities and identifying the most relevant ones for your teaching and AI focus
Running command cd /home/sandbox && python3 -c " import pandas as pd import re # ...
Agent is running...
You can step in with instructions while the task runs...
Live Activity
Outputs
Esc
Searched Gov grants faculty development medical school
id
number
title
agencyCode
agency
357,568
PAR-25-298
Research Enhancement Award Program (REAP) for Health Professional Schools and Graduate Schools (R15 Clinical Trial Not Allowed)
HHS-NIH11
National Institut
351,917
PAR-24-113
International Research Scientist Development Award (IRSDA) (K01) Independent Clinical Trial Not Allowed
HHS-NIH11
National Institut
351,871
PAR-24-114
International Research Scientist Development Award (IRSDA) (K01 Independent Clinical Trial Required)
HHS-NIH11
National Institut
357,632
PAR-25-196
Limited Competition: Mentored Research Career Development Program Award in Clinical and Translational Science Awards (CTSA) Program (K12 Clinical Trial Optional)
HHS-NIH11
National Institut
352,285
PAR-24-117
The NCI Transition Career Development Award (K22 Independent Clinical Trial Optional)
HHS-NIH11
National Institut
360,613
PAR-26-031
Centers of Biomedical Research Excellence: Expansion/Sustainability Phases (COBRE E/S)
HHS-NIH11
National Institut
352,810
PAR-24-128
Medical Scientist Training Program (MSTP) (T32)
HHS-NIH11
National Institut

Research Grants For Professors   New Chat   Share   Pricing

as a medical school professor in Long Island, New York, focusing on teaching grants and AI education for medical students.

Key Findings:

Database Search Results:

- **Teaching grants search:** 100 results
- **AI medical grants search:** 100 results
- **NIH/HRSA grants search:** 100 results
- **NSF education innovation grants:** 50 results

Top 3 Priority Opportunities:

1. 🏆 **NSF Digital Twins in Biomedical Innovation (24-561)**
   - $5M total funding, deadline May 4, 2026
   - Perfect fit for AI education in medical curriculum
   - Supports interdisciplinary research and education
2. ⚠️ **HRSA Teaching Health Center Graduate Medical Education (HRSA-26-011)**
   - $47.6M total funding, **deadline TODAY (Oct 20, 2025)**
   - Only for current THCGME recipients

Ask anything or give follow up task...

627   Deep Search

Live Activity   Outputs   Esc

Search Files

Final Outputs
- spring_fall_2026_grants_re...
- grants_spring_fall_2026.csv
- medical_school_grants_re...
- filtered_relevant_grants.csv

All Files

Download All Files

grants_spring_fall_2026.csv

id	number	title	agencyCode
353,135	24-561	Foundations for Digital Twins as Catalyzers of Biomedical Technological Innovation	NSF
349,692	PAR-23-240	Alcohol and Other Substance Use Research Education Programs for Health Professionals (R25 Clinical Trial Not Allowed)	HHS-NIH11
350,485	PAR-23-228	National Institute of General Medical Sciences Predoctoral Basic Biomedical Sciences Research Training Program (T32)	HHS-NIH11
352,282	PAR-24-127	NIDCD's Mentored Research Education Pathway for Otolaryngology Residents and Medical Students (R25 - Clinical Trial Not Allowed)	HHS-NIH11
351,947	24-529	Innovations in Graduate Education (IGE) Program	NSF
356,250	PAR-24-281	NLM Information Resource Grants to Reduce Health Disparities and Promote Health for All (G08 Clinical Trial Not Allowed)	HHS-NIH11
352,637	RFA-OD-24-013	Building Interdisciplinary Research Careers in Women's Health (BIRCWH) (K12 Clinical Trial Optional)	HHS-NIH11
355,177	PAR-24-233	NIMH Research Education Mentoring Program for HIV Researchers (R25 Clinical Trial Not Allowed)	HHS-NIH11
355,297	RFA-CA-24-026	National Cancer Institute Youth Enjoy Science Research Education Program (R25 Clinical Trial Not Allowed)	HHS-NIH11
355,492	PAR-24-252	Innovative Programs to Enhance Research Training (IPERT) (R25 Independent Clinical Trial Not	HHS-NIH11

ChatGPT

What can I help with?

filtered_relevant_grants.csv
Spreadsheet

Can you help me filter these grants to the ones most relevant to creating teaching modules to medical students on how to use AI for research? Also only grant with a deadline in spring 2026 or later

Relevance_Why	Parsed_Deadline	id	number	title	agencyCode	agency	openDate	closeDate	oppStatus	docType	cfdaList	relevance_score
audience match; AI match;	8/11/2026	353962	24-570	Research Training Groups in the Mathematical Sciences	NSF	U.S. National Science	5/3/2024	8/11/2026	posted	synopsis	['47.049']	2
audience match; AI match; research match	9/25/2026	350485	PAR-23-228	National Institute of General Medical Sciences Predoctoral Basic Biomedical Sciences Research	HHS-NIH11	National Institutes of Health	10/5/2023	9/25/2026	posted	synopsis	['93.859']	6

This opportunity appears aligned with creating educational modules that teach medical students how to use AI for research workflows. It is offered by U.S. National Science Foundation. Scope explicitly touches AI/ML/data science, aligning with your AI-for-research emphasis. Content overlaps research skills (literature, IRB, analysis, reproducibility), fitting your module outcomes.

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HUMANITIES GRANTS

AMANDA GOLDEN, PH.D.
ASSOCIATE PROFESSOR OF ENGLISH

FUNDING SOURCES

National Endowment for the Humanities

ACLS, Mellon, Private Foundations

Archives and Site-Specific Resources

Residential Fellowships (Year-Long Humanities Center fellowships, National Humanities Center)

Professional Organizations (Conference Travel Grants, Research Grants)

Tuition Scholarships for Programs (i.e., Digital Humanities Summer Institute)

SUGGESTIONS

- Review Grant Instructions and Evaluation Criteria
- Watch Videos of Sessions Led by Grant Administrators
- Read Sample Proposals
- Read Lists of Previously Funded Projects
- Ask Letter Writers Who Know Your Project Well
- For Some Fellowships, NEH Officers Will Read Drafts

SAMPLE REVIEW CRITERIA

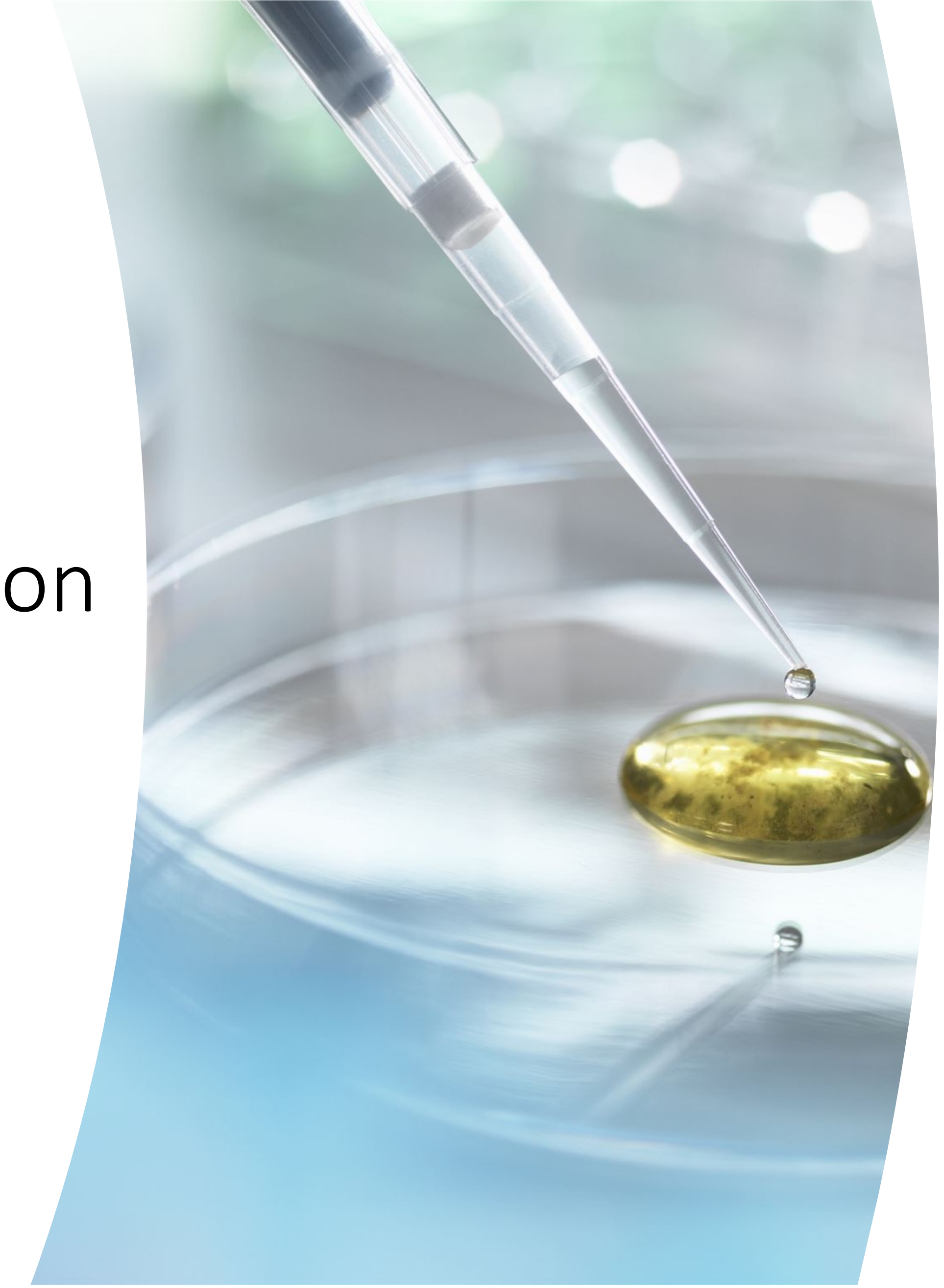
1. Review Criteria

Peer reviewers will use the following criteria to review applications under this notice:

1. The intellectual significance of the proposed project, including its value to humanities scholars, general audiences, or both.
2. The quality of the conception, definition, organization, and description of the project and the applicant's clarity of expression.
3. The feasibility and appropriateness of the proposed work plan.
4. The quality or promise of quality of the applicant as an interpreter of the humanities.
5. The likelihood that the applicant will complete the project (not necessarily during the period of performance), including, when relevant, the soundness of the dissemination and access plans.

Each review criteria corresponds to specific sections of the narrative and the other application components. See D2. Content and Form of Application Submission.

Introduction to NIH Funding



- **Largest Biomedical Funder**
  - https://grants.nih.gov/
- Funding level determined by congress annually
  - Cycle: Oct 1 – Sep 30
  - Invested ~$48 billion in FY2025, slight increase in FY2026
- 27 institutes and centers (ICs):
  - https://www.nih.gov/institutes-nih/list-institutes-centers
  - Budget level:
    - NCI (National Cancer Institute): $7.4 Billion
    - NIAID (National Institute of Allergy and Infectious Diseases): $6.5 Billion
    - NIGMS (National Institute of General Medical Sciences): $5.77 Billion

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NIH Funding Mechanisms

- **Research grants: R series** - research projects
- **Career Support: K series** - help researchers transition to independence
  - K01: mentored biomedical research
  - K08: clinical research
- **Training support: T and F series -** train biomedical researchers
  - T32: institutional programs with coursework, mentoring, and research experience
  - F31/F32 fellowships: provide stipend and training support for graduate and postdoc researchers
- **Small Business Initiatives: SBIR/STTR-**commercialization of intellectual properties
  - SBIR (Small Business Innovation Research): business at least 67% for Phase I, 50% for Phase II, no requirement of partnering with research institution
  - STTR (Small Business Technology Transfer): business at least 40%, research institution at least 30%, remaining 30% either
  - Much higher paylines
- **Supplemental funding:**
  - **Administrative supplements**: e.g., extra support for minority graduate students; does not require scientific peer review
  - **Competing revisions**: expands project scope, requires scientific peer review

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NIH Research Grants (R Series)

Types of R Series Grants

- **R01** supports major projects
  - Up to 5 years
  - Typically up to $500K/year DC (direct costs), higher with prior approval
- **R03** funds small pilot studies
  - Do not requires preliminary data
  - Up to 2 years
  - Up to $50K/year DC
- **R21** encourages innovative research: high risk/high reward
  - Do not require preliminary data, but including some strengthens the application
  - Up to 2 years
  - Up to $275K total DC
- **R15 (AREA/REAP):** AREA (Academic Research Enhancement Award) for undergrad-focused schools; REAP (Research Enhancement Award Program) for Health Professional Schools and Graduate Schools
  - Only institutions receiving < 6 million NIH funding (DC+IDC)/year during the last 4 of 7 FYs are eligible
  - Each PI may not hold an active NIH research grant or more than 1 R15 award at a time
  - Training students and strengthening research environment at the institutions
  - Up to 3 years
  - Up to $375K total DC ($125K/year)
  - 23 ICs participating: **NIGMS** – the most (33%) R15 grants; NICHD and NCI 8% each.

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New Investigator Advantage

- **New Investigator (NI)**:
    - A PI who has **not** previously competed successfully for a substantial, independent NIH research award (e.g., R01 or equivalent).
- **Early-Stage Investigator (ESI)**:
    - A **subset of NI**
    - A PI who qualifies as a New Investigator **and** is within **10 years** of completing their **terminal research degree** or **post-graduate clinical training** (whichever is later).
- Notes:
    - Much higher paylines
    - Reviewers are instructed to focus more on the proposal approach rather than PI's track record. Less preliminary data are expected.
    - Benefits generally applicable to R01 or equivalent applications

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine | Arkansas

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NIH Simplified Peer Review Framework

Previously 5 scored criteria: Significance, Investigator, Innovation, Approach, Environment, all scored 1 - 9

As of January 25, **2025**, most NIH Research Project Grant applications (e.g., R01, R03, R15, R21, U01) are reviewed using **three core factors** (https://grants.nih.gov/policy-and-compliance/policy-topics/peer-review/simplifying-review/framework)**:**

- **Factor 1: Importance of the Research**
    - Combines **Significance** and **Innovation**
    - Scored 1–9
    - Assesses how important and novel the proposed research is
- **Factor 2: Rigor and Feasibility**
    - Based on **Approach**
    - Scored **1–9**
    - Evaluates the scientific rigor, clarity, and feasibility of the methods
- **Factor 3: Expertise and Resources**
    - Combines **Investigator** and **Environment**
    - **Not scored numerically**; reviewers select from:
        - **Appropriate** (no explanation needed)
        - **Needs additional expertise/resources** (requires brief justification)
- All three factors contribute to the **Overall Impact Score**, which remains the primary metric for funding decisions.

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Resources and Tools

- NIH Grants & Funding Home:
  - https://grants.nih.gov
- Explore NIH Grant Opportunities:
  - https://grants.nih.gov/funding/explore-nih-opportunities
- NIH Guide for Grants and Contracts:
  - https://grants.nih.gov/grants/guide
- NIH **RePORTER** – search funded projects:
  - https://report.nih.gov
- NIH **eRA Commons** – track the status of the application:
  - https://public.era.nih.gov

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NSF Proposals

- PAPPG: NSF Proposal & Award Policies & Procedures Guide
- Requests for Proposals: Find, read and address!
- Merit Review Criteria
  - **Intellectual Merit:** The potential for the proposed project to advance knowledge and understanding within its own field or across different fields.
  - **Broader Impacts:** The potential for the proposed project to benefit society and contribute to the achievement of specific, desired societal outcomes.

Intellectual Merit essential for success!

Broader impacts may help push you over line.

Broader impacts should be unique to you!

DOE Visiting Fellow Program

- 10-week collaborative experience during the summer.
- Can invite up to two students
- NYIT faculty are currently eligible
- **VFP Research Collaboration track:**
  - Increase research competitiveness
  - Elevate your students' learning and research
  - Need a partner at a DOE lab
- **VFP Teaching Initiative track:**
  - Invigorates your STEM teaching through research collaboration with a DOE lab.
- Early Jan Deadline
- https://science.osti.gov/wdts/vfp

Writing A Compelling Proposal

Grant writing is different from paper writing!

- Obtain preliminary results/data
- Write persuasively & tell a story
  - State the need
  - Sell yourself and your research
  - Tie your project to funder's foci
- Write to the Merit Review Criteria
- Organization is key
  - Be clear about proposed vs prior work
  - Embed an outline review report in your proposal
- Initial draft can take a month of intensive work.
  - Set aside large blocks of time
  - Edit & proof read
  - Get comments

No go solo

- Find and study previous successful grant proposals.
- Discuss ideas with colleagues in the "same" field.
- Find colleagues willing to read and reread your proposal.
  - Talk to them about their comments.
  - Pay attention to what they failed to understand. Revise.
  - Get more comments. Revise. Repeat ....
- Consider writing collaborative proposals
- Get started with an NYIT ISRC grant
- NYIT OSPAR can help to identify funding sources, advise on proposal preparation, promote interdisciplinary teams.

If you don't succeed the first time,

revise and submit (again and again)!

Some Online Sources We Drew From

- Art of Grantsmanship
- GEO@NSF
- Compelling NSF Grant Proposal
- NSF: Preparing proposals
- Advice from CS Prof at CMU
- University of Oregon

Discussion Time!

Questions and Comments?